Tight-BDT Lower-Bound Crosscheck — Leakage-Corrected ABCD Purity

Nominal parametric tight cut vs. flat BDT > 0.70/0.80/0.85/0.90/0.95

PPG12 internal note

2026-06-01

Summary

To probe the sensitivity of the ABCD leakage-corrected purity to the tight-region BDT lower bound, the flat tight-cut crosscheck series (BDT > 0.70/0.80/0.85/0.90/0.95) was built out by adding the BDT > 0.80 and BDT > 0.70 points and running them through the full per-sample TTree pipeline. The leakage-corrected purity rises monotonically with the tight cut across the full range, exactly as expected. The nominal parametric cut is equivalent to a flat ≈ 0.80 threshold, and the loosest BDT > 0.70 point sits below the nominal at every bin. The comparison is robust. Two pre-existing config-hygiene issues in the flat-variant series were found, and neither biases this comparison.

1 Method

ABCD regions. The four regions are A = tight & isolated (signal region), B = tight & non-isolated, C = non-tight & isolated, and D = non-tight & non-isolated. The combinatorial background in A is estimated as $R\,(B\,C)/D$. The leakage-corrected purity is

$$P_{\text{leak}} \;=\; \frac{N^A_{\text{sig}}}{N_A},$$

solved self-consistently with the signal-MC leakage fractions

$$c_B, c_C, c_D \;=\; \frac{\text{signal MC in } B, C, D}{\text{signal MC in } A},$$

as implemented in `efficiencytool/CalculatePhotonYield.C`. The result is stored as the `TGraphErrors` `gpurity_leak` in `Photon_final_{var}.root`.

Nominal tight cut. The nominal tight boundary is parametric in transverse energy,

$$b_{\min}(E_T) \;=\; 0.815625 \;-\; 0.0015625 \cdot E_T,$$

i.e. about 0.80 at $E_T = 10\,\text{GeV}$ and 0.76 at $E_T = 36\,\text{GeV}$.

Flat crosscheck variants. Each flat variant sets the tight region to BDT > T (flat in E_T) and the non-tight upper bound to T (no gap at the tight boundary), for $T = 0.70, 0.80, 0.85, 0.90, 0.95$.

Pipeline. All variants were produced with the full per-sample TTree-reader flow. Phase 1 (HTCondor) ran `RecoEffCalculator_TTreeReader.C` on all 8 SI/DI MC sample pairs plus data per crossing-angle period, followed by `MergeSim` and per-period `CalculatePhotonYield`. Phase 2 merged the periods (lumi-weighted hadd) and ran the all-range `CalculatePhotonYield`. All results are all-range, all-z nominal ($64.3718\,\text{pb}^{-1}$), double-interaction-blended (0.224 at 0 mrad, 0.079 at 1.5 mrad), with truth-vertex reweighting on.

Provenance. The BDT > 0.85/0.90/0.95 points already existed (produced 2026-05-14) and were verified current and sane, then reused. The BDT > 0.80 (`flat_t80_nt50`) and BDT > 0.70 (`flat_t70_nt50`) points were newly added: each config was mirrored from the `t85` config, changing only `var_type` and the two flat-cut intercepts (0.85 → 0.80 and 0.85 → 0.70), and run end-to-end on 2026-05-31 and 2026-06-01 respectively.

2 Results

Table 1: Leakage-corrected purity (`gpurity_leak`) per E_T^{γ} bin center [GeV], all-range $p+p$ at $\sqrt{s} = 200\,\mathrm{GeV}$, $64.37\,\mathrm{pb}^{-1}$. The BDT > 0.95 series exhausts its last bin at $E_T = 34\,\mathrm{GeV}$ (entry shown empty).

E_T [GeV]	nominal	BDT > 0.70	BDT > 0.80	BDT > 0.85	BDT > 0.90	BDT > 0.95
11	0.511	0.473	0.512	0.541	0.583	0.664
13	0.563	0.527	0.559	0.594	0.655	0.774
15	0.624	0.586	0.614	0.647	0.695	0.801
17	0.727	0.691	0.710	0.740	0.768	0.866
19	0.697	0.666	0.671	0.729	0.798	0.930
21	0.656	0.674	0.669	0.671	0.744	0.870
23	0.871	0.848	0.886	0.929	0.963	1.000
25	0.877	0.817	0.819	0.876	0.912	0.787
27	0.828	0.634	0.889	0.835	0.879	0.924
30	0.730	0.652	0.630	0.607	0.743	0.947
34	0.983	0.972	0.960	0.992	1.010	—

Table 2: Region-A (tight & isolated) data yield, physical-bin integral, all-range, ordered loose to tight.

selection	region-A yield
BDT > 0.70	96039
nominal	84156
BDT > 0.80	83625
BDT > 0.85	75182
BDT > 0.90	62744
BDT > 0.95	37464

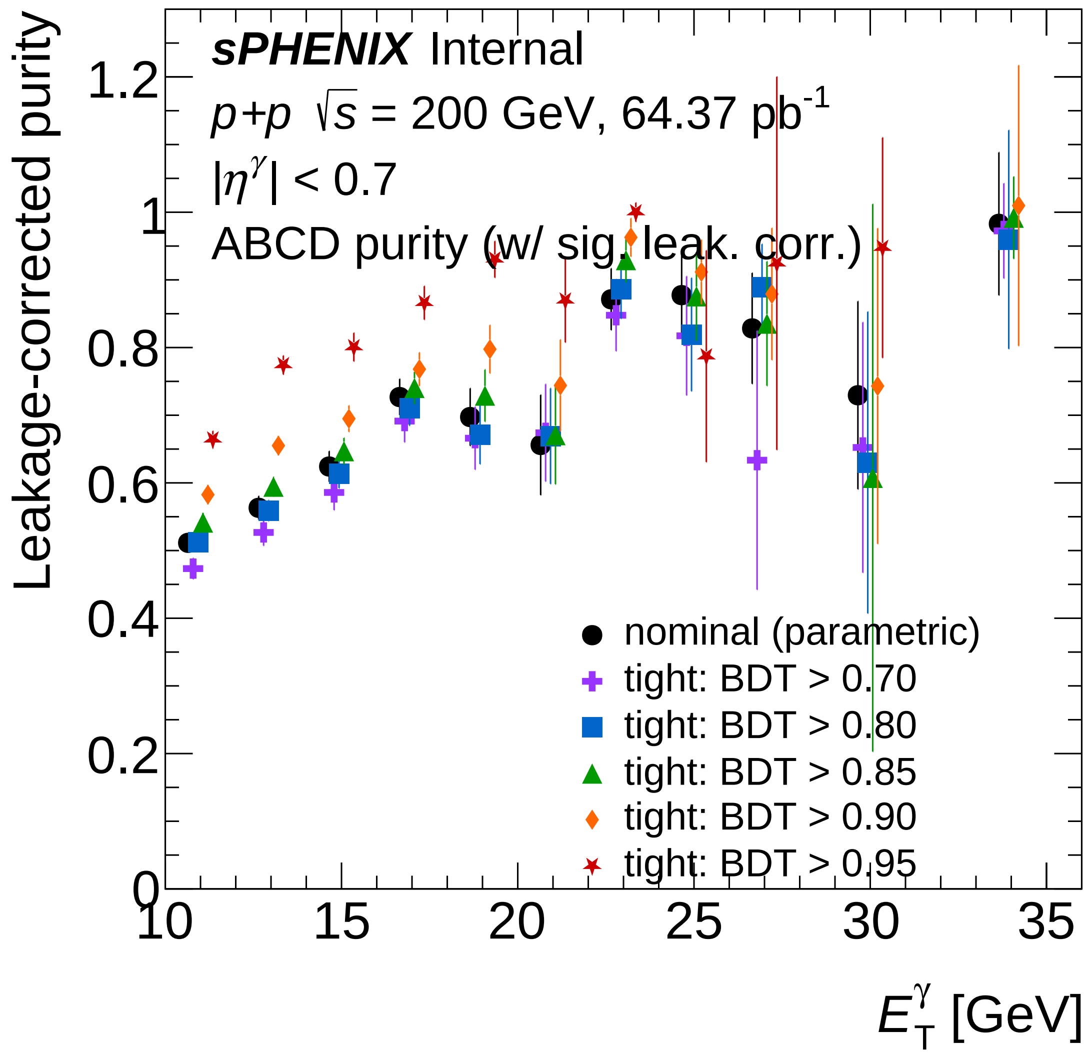

Figure 1: Leakage-corrected ABCD purity vs. E_T^γ for the nominal parametric tight-BDT cut and the flat tight cuts BDT $> 0.70/0.80/0.85/0.90/0.95$, all-range $p+p$ at $\sqrt{s} = 200\,\mathrm{GeV}$, $64.37\,\mathrm{pb}^{-1}$. Points are offset horizontally by up to $0.35\,\mathrm{GeV}$ for clarity. No smooth-fit overlay is drawn (points with vertical errors only).

Reading the result.

- **Monotonic rise with the tight cut.** The purity increases with the tight cut at every well-measured (low-to-mid E_T^γ) bin, across the full range BDT $> 0.70 \to 0.95$. For example at $E_T = 15\,\mathrm{GeV}$ the purity goes $0.59 \to 0.61 \to 0.65 \to 0.70 \to 0.80$ for BDT $>$ $0.70/0.80/0.85/0.90/0.95$. The loosest cut 0.70 sits below the nominal at every bin, and the tightest 0.95 gains roughly $+0.15$ to $+0.25$ absolute purity over the nominal/0.80 at low-mid E_T^γ.
- **Steep statistics-vs-purity trade.** Tightening from the loosest 0.70 to 0.95 discards a factor $\sim$2.6 of region A (96k $\to$ 37k clusters, Table 2), which is why the 0.95 series is noisy at high E_T^γ.
- **Nominal $\equiv$ flat $\approx$ 0.80.** The BDT > 0.80 variant has nearly the same region-A yield as nominal (83625 vs. 84156, 0.6% lower) and tracks the nominal purity within the toy uncertainties. The 0.6% region-A yield difference reflects the tight-boundary shape (nominal parametric, slightly looser than flat 0.80 at high E_T^γ), and the small residual *purity* differences reflect the flat no-gap non-tight topology vs. the nominal tight/non-tight gap.
- **High-E_T^γ is statistics-limited.** Above $\sim 24\,\mathrm{GeV}$ the comparison is dominated by toy un-

certainties (large error bars, occasional ABCD overshoot above 1). The BDT > 0.95 series exhausts its last bin at $E_T = 34\,\mathrm{GeV}$ (shown empty). The figure shows points with vertical errors only, with no smooth-fit overlay.

3 Findings and caveats

Table 3: Findings from the flat-variant series. None biases the tight-cut purity comparison reported here.

severity	finding
WARNING	**`nt_bdt_min` override is shadowed.** The flat configs set `non_tight.bdt_min` (the flat non-tight floor labeled "`nt50`" = 0.50), but `RecoEffCalculator_TTreeReader.C` line 572 reads the inherited parametric `non_tight.bdt_min_intercept` (0.7333) $-\,0.0133 \cdot E_T$, which takes precedence. So the non-tight floor is the nominal parametric value in every flat variant, never the labeled flat 0.50. Consequence: `flat_t90_nt50` and `flat_t90_nt70` are numerically identical (verified: region A, B, C, D yields and `gpurity_leak` match to all digits). This does *not* bias the tight-cut purity comparison here, because the shadowed non-tight floor is identical (common-mode) across all of BDT > 0.70/0.80/0.85/0.90/0.95 and cannot drive the trend. The intended per-variant change is the shared tight floor and non-tight ceiling at T. It only makes the "`nt50`/`nt70`" labels misleading. *Recommendation:* if a genuine flat non-tight floor is wanted, the variant generator must also set `non_tight.bdt_min_slope = 0` and `non_tight.bdt_min_intercept` to the flat value, not just `non_tight.bdt_min`.
INFO	BDT > 0.95 **Padé purity fit degenerates** on the near-unity points at high E_T^{γ} (the stored `f_purity_leak_fit` is unreliable for 0.95). The crosscheck figure uses points only, so this does not affect it.
INFO	BDT > 0.80 ≈ **nominal,** BDT > 0.70 **below it.** The flat 0.80 cut equals the nominal parametric cut to better than 1% in region-A yield, and the flat 0.70 cut is looser than nominal everywhere (region-A yield 96039, the largest of the series).

Reviewer pass matrix

Artifact	2a physics	2b cosmetics	2c re-derivation	3.5 fix-validator
`flat_t80`/`flat_t70` config + pipeline	PASS (comparison valid, `nt_min` bug flagged)	N/A	PASS (t80, t70 purity + A-count confirmed)	PASS (Phase 1+2 rc=0, A=83625/96039 sane)
purity comparison figure	N/A	PASS (deploy gate)	PASS	N/A
existing t85/t90/t95	PASS	shown in figure	PASS (all confirmed)	N/A

Files

New.

- `efficiencytool/config_bdt_flat_t{80,70}_nt50{,_0rad,_1p5mrad}.yaml`
- `efficiencytool/flat_t{80,70}_feeders.list`, `flat_t{80,70}_phase1.sub`
- `plotting/plot_purity_tightbdt_crosscheck.C`
- `plotting/figures/purity_tightbdt_crosscheck.pdf`
- `efficiencytool/results/Photon_final_bdt_flat_t{80,70}_nt50{,_0rad,_1p5mrad}{,_mc}.root`

Modified.

- `efficiencytool/make_bdt_variations.py` (added `flat_t80_nt50` and `flat_t70_nt50` variants)
- `efficiencytool/all_feeder_configs.list` and `all_range_configs.list` (added `t80`, `t70` entries)